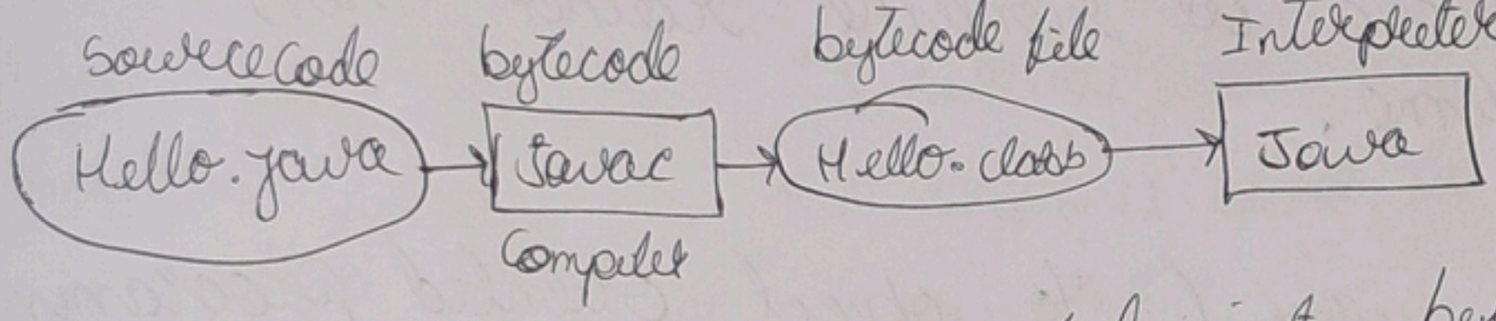
25/09/26

ASSIGNMENT-1

① Explain how java achieves platform independence through the concepts of Bytecode and JVM.

A) Java is an interpreted language and a platform independant language which ensures that it is capable of running on any system which has JVM in it which is abbreviated as Java Virtual Machine.



- The source code is converted into bytecode by the compiler which is then interpreted by the virtual Machine which is JVM.

- Java bytecodes are machine independent which means that they will be created on a system and the bytecode is transferred to other machine and is executed with JVM on that machine.

- JVM is a virtual machine that executes the bytecode which contains of a set of formatted instructions of what the machine has to perform.

- JVM is an abstract machine that is implemented by emulating it in software on a real machine.

(2) What do you mean by dynamic binding? How is it useful to object oriented programming.

A) Dynamic binding is defined as the programming practice which decides which function or method is to be executed on being called and a best example of this dynamic binding is method overriding which will be done at runtime rather than compile time.

This is useful to object oriented programming at the time of inheritance where the methods called will be executed and dynamic binding helps in code reusability because a method or function which is written once can be inherited and can be used in another class (or) method as per the requirements.

This dynamic binding on the other hand enables and helps with polymorphism and helps to decode extensible Code.

(3) Discuss the differences in programming among procedural languages and object oriented languages.

A) PROCEDURAL LANGUAGES

- these mainly focus on the methods ie how to perform a task.
- They follow a top down approach, ie starts with main problem and then breaks them into sub-problems.
- Large programs are divided into functions
- there is no data security, as the data moves freely in the system and the access specifiers are not supported.
- there is very limited scope for code reusability.

OBJECT ORIENTED LANGUAGES

- these mainly focus on how the data is being modified.
- They follow a bottom-up approach where objects are designed first.
- programs are divided into modules called as objects.
- high security as the data is bound to the functions by encapsulation.
- The code reusability is very high.

④ Describe inheritance as applied to Object oriented programming

A) Inheritance is the process of allowing the features of the parent class to a child class in a hierarchical structure. Using inheritance we can create a base class that has the common features between all of them. (or) the classes that are about to be created where they can be inherited.

The class that is inherited is called a superclass while the class which is inheriting is called a sub class. The different types of inheritance are multilevel inheritance there is an hierarchical order.



- Inheritance allows method overriding by which allows java to support run-time polymorphism. This allows either to add or modify the functionalities of the methods that are inherited.

(5) What do you mean by just In Time (JIT) compilation?

A) JIT is a technique, where the program's code is compiled at the time of running into a byte code rather than being entirely compiled before execution and because of this, the performance of the system (or) the code might increase, because the compiler focuses on the key aspects of the code. this produces a bytecode at the time of ~~compilation~~ run time and not before that as the way it happens in some traditional compilers. This helps even in optimization during runtime behaviour.

Ex:- Let us consider there are functions add(), sub() & multiply() and if a code executes and the add() function runs for a large number of times, then the code in it is optimized as that helps in faster performance.

⑥ Discuss various features of Java.

A) The features of Java are:-

① Simple

- It is a language which has no pointers involved and can be easily typed and it is easy to understand.

② Secure

- It is secure as it keeps inbuilt JVM which is more like a simulation software which is free of all the viruses and as applets are not allowed to access the files. The security is high.

③ Portable

- Considered to be a key feature and this is considered because the code can be compiled on a machine and then the byte code can be transferred onto a different machine and can even run on that machine if that machine has JVM in it.

④ Object Oriented

- Java follows OOP's principles and because of that the code reusability is enhanced and even the features of OOP's are used by Java.

⑥ Robust

- It is robust language it is strong language and exception handling is compulsory due to which bugs can be found safely and can be resolved and the errors can be descriptive.
- Automatic garbage collection during the program which make sures that the created variables are taken the space off so as to save the space.